

AVR 8-BIT HEX FILE MANIPULATOR

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Various disassemblers, including the AVR Studio from Atmel, are available for free download from the Internet. AVR Studio, in particular, can easily disassemble the hex file of AVR microcontrollers. The disassembler details the data code and its memory location along with the program instruction code.

Here, we describe a manipulator for AVR hex files that displays:

1. Hex file in '.hex' format
2. Hex file in decoded Intel 8-bit hex format
3. High and low data on each address space

The program is written in Visual C# on the .NET platform that comes along with Microsoft Visual Basic Studio 2008. It is designed to use Intel 8-bit hex format files for AVR series of microcontrollers. The software is a first step towards the development of a customised programmer for AVR microcontrollers that employs encrypted bootloaders.

Typically, hex files have the data structure containing the instruction code, start of file, end of file, number of bytes, address, etc. This software extracts the instruction code along with its location address for an unknown hex file. It is also useful to examine the record format at each line of the program code and manipulate the desired data.

After understanding the programming code, you can develop your own software for reading/writing hex files on the screen in any style. The study of the software will reveal the technique of data decoding from Intel 8-bit hex file format. You

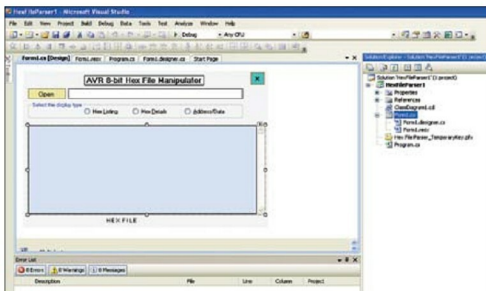


Fig. 1: Screenshot of the program development stage

can run the hex file parser program just by installing the .NET 2.0 file as explained in 'Making standalone application' section on page 93. The .NET 2.0 file is included in the EFY-CD almost every month under 'Resources' folder.

Intel 8-bit hex file format

Defined by Intel, hexadecimal object file format is a way of representing an absolute binary object file in ASCII format. This means that the binary data in a hex file is stored in ASCII format, so it can be stored in tapes, line printers and punch cards and displayed on CRT terminals. For example, the 8-bit binary value 1000-1110 is '8E' in hexadecimal. This is encoded in ASCII as 0011-1000 (38h) for '8' and 0100-0101 (45h) for 'E.' Thus representation in hex file format requires twice as many bytes as the binary representation. You may refer to the encoding scheme of Intel hexadecimal object file format in the Intel Hexadecimal Object File Format Specification Revision A,

1/6/88, which is readily available on the Internet.

Software program

The logic of the software lies within the .dll file created for the Visual Basic project. Create a 'New Project' of Visual C#.NET by selecting 'Windows Forms Application' option in Microsoft Visual Studio 2008, wherein a blank form is displayed.

Place a button with 'Open' label on the form and associate 'openFileDialog' control with the form, which opens the traditional 'Open File' dialogue box. The 'Open' button control, along with 'openFileDialog' control, loads the required hex file in the program.

Place 'Hex Listing,' 'Hex Details' and 'Address/Data' radio buttons inside 'groupBox' control, so any one of these can be selected at a time. Clicking a radio button processes the HexParser.dll file to display the resultant content in text format in the textbox placed below the radio button. The screenshot of the program development stage is shown in Fig. 1.

